

Questions

1. An operation has a 10 percent scrap rate. As a result, 72 pieces per hour are produced. What is the potential increase in labor productivity that could be achieved by eliminating the scrap?
2. A company that makes shopping carts for supermarkets and other stores recently purchased some new equipment that reduces the labor content of the jobs needed to produce the shopping carts. Prior to buying the new equipment, the company used five workers, who produced an average of 80 carts per hour. Workers receive \$10 per hour, and the machine costs \$40 per hour. With the new equipment, it was possible to transfer one of the workers to another department, and the equipment cost increased by \$10 per hour while output increased by four carts per hour.
 - a) Compute labor productivity under each system. Use carts per worker per hour as the measure of labor productivity.
 - b) Compute the multifactor productivity under each system. Use carts per dollar cost (labor plus equipment) as the measure.
 - c) Comment on the changes in productivity according to the two measures, and on which one you believe is the more pertinent for this situation.
3. Compute the multifactor productivity measure for each of the weeks shown for the production of chocolate bars. What do the productivity figures suggest? Assume 40-hour weeks and an hourly wage of \$12. Overhead is 1.5 times the weekly labor cost. Material cost is \$6 per pound.

Week	Output (units)	Workers	Material (lbs)
1	30,000	6	450
2	33,600	7	470
3	32,200	7	460
4	35,400	8	480

4. A company offers ID theft protection using leads obtained from client banks. Three employees work 40 hours a week on the leads, at a pay rate of \$25 per hour per employee. Each employee identifies an average of 3,000 potential leads a week from a list of 5,000. An average of 4 percent actually sign up for the service, paying a one-time fee of \$70. Material costs are \$1,000 per week, and overhead costs are \$9,000 per week. Calculate the multifactor productivity for this operation in fees generated per dollar of input.